

Rock, Paper, Scissors Showdown



Write a Python program that lets you play Rock, Paper, Scissors against the computer.

STEPS

1. Ask the player to enter 1 for Rock, 2 for Paper, or 3 for Scissors.
2. Have the computer randomly choose 1, 2, or 3 of its own.
3. Print what the player chose and what the computer chose.
4. Compare the two choices: Rock beats Scissors, Scissors beats Paper, Paper beats Rock.
5. Print whether the player won, lost, or tied.

STRETCH GOAL

Let the player keep playing multiple rounds in a loop, and keep score across rounds.

Solution

HINTS

- Python's random module can pick a random number for you -- look up `random.randint()`.
- Use `int()` to convert the player's typed input into a number you can compare.
- Check for a tie first (both pick the same number) before checking who wins.

STEPS

1. Import the random module.
2. Ask for and store the player's choice as an integer.
3. Generate the computer's choice with `random.randint(1, 3)`.
4. Print both choices by name so the player can see what happened.
5. Use `if/elif/else` to check for a tie, then check the three ways the player can win, otherwise the computer wins.

```
import random

CHOICES = {1: "Rock", 2: "Paper", 3: "Scissors"}

player = int(input("Choose 1-Rock, 2-Paper, or 3-Scissors: "))
computer = random.randint(1, 3)

print(f"You chose {CHOICES[player]}. Computer chose {CHOICES[computer]}.")

if player == computer:
    print("It's a tie!")
elif (
    (player == 1 and computer == 3) or
    (player == 2 and computer == 1) or
    (player == 3 and computer == 2)
):
    print("You win!")
else:
    print("Computer wins!")
```